

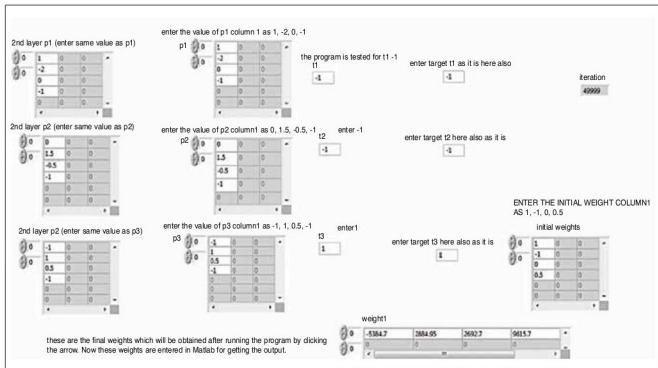


Fig. 3: Front panel of LabVIEW program for feeding the inputs

ing in neural nets is primarily a process of adjusting connection strengths. In neural nets of the type described so far, the most popular method of learning is called 'perceptron,' which is a basic algorithm to learn neural network theory.

The perceptron is a type of artificial neural network invented in 1957 at the Cornell Aeronautical Laboratory by Frank Rosenblatt. It can be seen as the simplest kind of feedforward neural network: a linear classifier. The perceptron is a binary classifier which maps its input 'x' (a real-valued vector) to an output value 'f(x)' (a single binary value):

$$f(x) = \begin{cases} 1 & \text{if } w \cdot x + b > 0 \\ 0 & \text{otherwise} \end{cases}$$

where 'w' is a vector of real-valued weights, 'w.x' dot product (which here computes a weighted sum) and 'b' bias (a constant term that does not depend on any input value).

The value of 'f(x)' (0 or 1) is used to classify 'x' as either a positive or a negative instance, in the case of a binary classification. If 'b' is negative, the weighted combination of inputs must produce a positive value greater

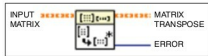


Fig. 4: Transpose matrix

than |b| in order to push the classifier neuron over the '0' threshold. Spatially, the bias alters the position of the decision boundary. The perceptron learning algorithm does not terminate if the learning set is not linearly separable.

In Fig. 1, p_1 and p_2 are the inputs to the system. These are implemented in LabVIEW. The inputs are then weighted with some suitable values such as w_{11} and w_{12} . Now the inputs 'p' and 'w' are multiplied and all the products ($p \times w$) are added. Since the values obtained after this step are random real numbers, they need to be converted into a binary number. This is obtained by using the hardlimit functions (refer Fig. 2) like unit step or signum function. After a number of iterations, the final weights are obtained.

In Figs 1 and 2, 'b' represents bias, 'n' number of types of inputs, 'R' real

number, and 'S' unit step function to convert the real number into 1's and 0's. $R \times 1$ and $S \times 1$ represent multiplication.

The final weights are computed to get the final output using MATLAB.

LabVIEW block description

LabVIEW is a graphical programming environment that allows you to design and analyse a digital processing system in a shorter time than text-based programming environments. LabVIEW graphical programs are called virtual instruments.

Virtual instruments run based on the concept of data flow programming. This means that execution of a block or a graphical component is dependent on the flow of data, or more specifically a block executes when data is available at all of its inputs. Output data of this block is then sent to all other connected blocks.

A virtual instrument consists of two major components: a front panel and a block diagram. The front panel provides the user-interface of a program, while the block diagram incorporates